NEWSBOT INTELLIGENCE SYSTEM

INDIVIDUAL TECHNICAL REPORT | ITAI 2373 NATURAL LANGUAGE PROCESSING

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GitHub: https://github.com/IdrisJunaidAI/ITAI2373_NewsBot_Midterm

Executive summary and business case

NewsBot is a reproducible natural language processing system for categorizing news articles and extracting interpretable signals for media monitoring and business intelligence. It combines cleaning, lexical features, grammatical and syntactic analysis, sentiment assessment, supervised classification, and named entity recognition in one Google Colab notebook. The instructor approved BBC News Classification corpus is the evaluation source. This is a research and education prototype rather than a production decision engine.

Dataset quality and processing

The approved BBC training file contained 1,490 labeled articles. After filtering the project retained 1,440 English news articles across five categories: sport (342), business (335), politics (266), entertainment (263), and technology (234). The pipeline validates its schema, handles missing values and duplicates, and applies normalization, tokenization, stop word filtering, and lemmatization. The BBC dataset does not contain a separately labeled Health category.

Integrated NLP architecture

Modules 1–3. Define the editorial and business intelligence use case, prepare textual inputs, and transform documents into TF IDF features with term level analysis.

Modules 4–5. Use spaCy for part of speech distributions, dependency parsing, and approximate grammatical relationship extraction.

Modules 6–8. Apply VADER sentiment and exploratory emotion cues; compare supervised multiclass models; extract people, organizations, places, dates, and monetary entities.

Evaluation design and model comparison

A stratified 80/20 train and test split was used, with three fold cross validation performed within the training portion. Vectorization was fitted inside each learning pipeline to reduce leakage. The best model was selected by cross validation macro F1, followed by evaluation on an untouched held out sample of 288 articles.

Model	CV accuracy	CV macro F1
Linear SVM	97.05%	96.94%
Logistic regression	96.27%	96.15%
Multinomial Naive Bayes	95.49%	95.34%
Majority class baseline	23.78%	7.69%

Held out Linear SVM accuracy: 96.53% | Macro F1: 96.45%

Results and interpretation

Classification reliability. Linear SVM outperformed the alternative models on training cross validation. On the held out set, technology recall was 0.89 (47 examples), while sport recall was 1.00 (68 examples). This identifies technology classification as a priority for further error analysis. Results apply to this BBC split and cannot establish equal performance on current publishers or new domains.

Grammatical patterns. Technology articles showed a higher noun share (0.261) than sport (0.208); sport showed a higher pronoun share (0.078) than business (0.039). These are descriptive differences in this sample, not evidence of causation.

Sentiment and entities. VADER assigned negative sentiment to 46.2% of politics articles versus 17.9% of entertainment articles. The notebook recorded 27,756 entity mentions and 7.263 extracted syntactic relations per article on average. Entity cooccurrence suggests relationships to investigate; it does not verify actual affiliations.

Technical quality assurance and individual contribution

My documented work included acquiring and validating the instructor approved BBC dataset, configuring and running the project in Google Colab, investigating incompatible NumPy and pandas installations, troubleshooting scikit learn import failures, addressing the blocked NLTK VADER lexicon download, and correcting a mistaken housing data upload. I reviewed the saved classification outputs and NLP visualizations, prepared the individual GitHub portfolio documentation, and interpreted business relevance. These are contributions I can substantiate through the execution and troubleshooting record; this report does not claim that I personally authored every original notebook component or performed work attributable to another team member.

Limitations, governance, and future development

The source corpus is historical and consists of five categories; there is no Health labeled test class. Lexicon sentiment is sensitive to irony, negation, and journalistic framing, while dependency and entity extraction can introduce false relations and mislabeled names. Classifier performance should be revalidated on fresh, diverse articles and checked for dataset shift. Recommended extensions are confidence based human review, category specific error audits, provenance metadata, model and dataset version controls, and a lightweight dashboard. In pharmaceutical quality and regulatory intelligence settings, such outputs should support qualified reviewers rather than replace validated decision processes.

Conclusion

The executed NewsBot system demonstrates a coherent NLP workflow from raw news text to classification and interpretable signals. Its 96.53% held out accuracy is encouraging within the selected BBC dataset, while the observed category level weaknesses and limits of sentiment and entity extraction reinforce the importance of validation, traceability, and human oversight.

References

Kaggle. BBC News Classification. https://www.kaggle.com/competitions/learn-ai-bbc/data

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